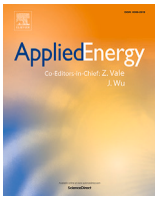
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Efficient power system corrective control with adaptive operation condition handling: A GATN-RL based method

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HIGHLIGHTS

- A novel GATN-RL framework explicitly models grid topology for corrective control.
- Temporal fusion mechanism enables proactive decisions under renewable fluctuations.
- Topology-aware embedding ensures robustness to dynamic network configurations.

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ABSTRACT

Traditional corrective control algorithms often struggle to strike a balance between computational accuracy and speed. While deep reinforcement learning (DRL) has emerged as a promising alternative, conventional DRL methods cannot capture the complex topological structures and temporal dynamics inherent to power systems. This paper proposes a corrective control algorithm based on a Graph Attention Temporal Network with Reinforcement Learning (GATN-RL) to address these challenges. The method employs a topology embedding mechanism that encodes connectivity directly into node features, enhancing the model's perception of the grid's structure. Specifically, Graph Attention Networks (GATs) are employed to aggregate data from adjacent nodes and capture spatial dependencies. A Transformer architecture is then used to identify temporal dynamics from sequential power system observations. Case studies on the 36-node and 118-node test systems validate that the proposed method delivers fast and effective corrective control. Compared to conventional reinforcement learning methods, GATN-RL improves the success rate by 2.7 % to 25 % in the 36-node system and by 9.83 % to 69.5 % in the 118-node system, while requiring less than 0.2 % of the computation time of traditional corrective control algorithms. Furthermore, the algorithm demonstrates strong generalization capabilities across diverse network topologies and under varying levels of renewable energy integration.

1. Introduction

Maintaining power flow within established operational limits is fundamental to ensuring secure and stable system operation. Under normal conditions, automatic generation control (AGC) effectively mitigates power fluctuations arising from load variations [1]. However, when contingencies such as scheduled maintenance or line faults result in line overloads, system operators must take corrective action. These overloads are typically resolved by re-dispatching generators, which is a critical task given the increasing integration of large-scale renewable energy sources and the need for enhanced system security [2]. In recent years, the integration of large-scale renewable energy sources and the deepening of reforms in power markets have contributed to increasing system uncertainties and complexities [3,4]. As a result, there have been more frequent instances of line overloads and faults, leading to power outages that can result in significant economic losses and disruptions to social activities [5,6]. To prevent such incidents, it is essential to develop

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Nomenclature

α_{ij} Normalized attention factor of node j to node i
β Sample factor
η Soft update factor
$\mathbf{A}_t$ Adjacency matrix at time t
$\mathbf{X}_t$ Feature matrix at time t
$\mathcal{N}(i)$ Neighborhood node set of i
Ω The set of all power equipment, including generators, loads, and both the original and extremity points of connecting lines
Ω_L The set of all lines
$\overline{\rho}_i$ Line load ratio threshold of node i
$\rho_{\max,t}$ Maximum line load ratio at time t
$\rho_{i,t}$ Load ratio of line i at time t
$\sigma(\cdot)$ Sigmoid function
θ, θ' The Q-network and target Q network parameters
θ^S, θ^T Parameters of GATN model and temporal model
$C(\cdot)$ Linear cost function
C_P, C_A Unit cost of re-dispatching active power of generators and bus splitting actions
c_t Connection state of electrical equipment at time t
D, D_{demo} Replay buffer and demonstration replay buffer
e_{ij} Correlation coefficient of nodes i and j
G_t Graph state at time t
$h_{\text{output},t}$ Output features of GAT at time t
$h_{i,t}, h_j$ New and old features of node i
$L(\theta)$ Loss function
M Batch size
N_G The number of generators
N_L The number of transmission lines
N_{demo} The size of demonstration replay buffer
N_{topo} The topology of power systems
$P_{i,\text{Gmin}}, P_{i,\text{Gmax}}$ Lower and upper bounds of generator i outputs
$P^E_{i,t}, Q^E_{i,t}$ Active and reactive power of line extremity i at time t
$P^G_{i,t}, Q^G_{i,t}$ Active and reactive power of generator i at time t
$P^L_{i,t}, Q^L_{i,t}$ Active and reactive power of load i at time t
$P^O_{i,t}, Q^O_{i,t}$ Active and reactive power of line origin i at time t
P_{Li} Transmission active power of line i
Q, V, A Q value, value function and advantage function
$r_{\rho,t}$ Reward for line overload at time t
$r_{\text{cost},t}$ Reward for action cost at time t
$R_{\text{down}}, R_{\text{up}}$ Lower and upper bounds of ramp capabilities
S, A, R, P, γ State space, action space, reward, transition matrix and discount factor
s_t, a_t, r_t State, action and reward at time t
T Operation time of power systems
u_B Action vector of splitting buses
u_G Action vector of re-dispatching generators
V^t_i Voltage at node i and time t
$V_{\min}, V_{\max}$ Minimum and maximum voltage bounds
W Weights of neural networks
y_t Target value
Z_t The output vector of the Transformer at time t

efficient corrective control methods when line overloads occur, thereby ensuring the stable operation of power systems [7].

Power system corrective control seeks to eliminate line overloads by adjusting generator outputs or modifying the network topology to redistribute power flow [8]. Traditional corrective control methods are broadly categorized into optimization-based algorithms and sensitivity-based algorithms. Optimization-based algorithms formulate complex mixed-integer nonlinear programming (MINLP) problems, which are then solved using mathematical programming algorithms [9]. While these methods achieve high accuracy, they are computationally expensive. Consequently, research has focused on simplifying the models or developing more efficient solution algorithms. For instance, many studies have adopted Mixed-Integer Linear Programming (MILP) formulations, which effectively handle discrete decision variables within a linear framework [10,11]. Other works employ second-order cone programming (SOCP) to establish robust stability constraints, enabling efficient convex approximations of the nonlinear stability region [12,13]. Heuristic algorithms have also been proposed to reduce the solution space and accelerate computation [14]. However, these conventional methods have several significant limitations. First, they depend on precise and highly complex physical models, which demand substantial computational time [15]. Although considerable research has aimed to reduce computational complexity, the solution time still tends to grow exponentially with the power system's scale, limiting their applicability to large-scale networks [16]. Second, in power systems with high renewable energy penetration, fluctuations from generation sources and loads cause frequent changes in operating conditions. This volatility can increase the time required for algorithms to converge and may even lead to model infeasibility [17]. Third, the solutions generated by these methods often require simultaneous adjustments to numerous generating units, which complicates their practical implementation in real-world systems [18].

Sensitivity algorithms determine the adjustment sequence for generators by ranking them based on their influence on transmission line power flows. A common approach in the literature is to re-dispatch generators according to their power flow sensitivity indices [19]. To better reflect system dynamics, multi-machine sensitivity analysis methods have been proposed to incorporate quasi-steady-state physical responses [20]. Other research has integrated flexible control devices and developed dynamic search algorithms that use power sensitivity to select control variables for corrective actions [21]. However, these sensitivity-based approaches have significant limitations. While effective at mitigating a single line overload, they tend to overlook the loading conditions on other lines. This oversight can trigger new overloads elsewhere in the system, necessitating repetitive adjustments. Moreover, the application of sensitivity algorithms is typically restricted to generator re-dispatching and cannot be applied to topology adjustments, which significantly limits their operational flexibility [22]. In summary, traditional corrective control methods, constrained by their reliance on precise models and complex computations, fail to balance efficiency and accuracy. This makes them increasingly inadequate for the complex demands of modern power systems with high renewable energy penetration, creating an urgent need for new, data-driven paradigms.

Artificial intelligence (AI) technologies, especially deep reinforcement learning (DRL), have opened new research opportunities in power system corrective control due to their strong perception and decision-making abilities [23,24]. DRL works by maximizing rewards through continuous interactions between environments and agents [25]. In recent years, researchers have utilized DRL to solve power system control problems. A multi-objective DRL reward design method has been proposed for power flow optimization [26]. A simulation-driven strategy based on DRL has been proposed for corrective control [27]. Some researchers combine the sensitivity algorithm, employing sensitivity factors to guide the DRL agent in managing line overloads [28]. By utilizing neural networks to construct agents, DRL is less affected by the scale and complexity of systems. It possesses strong nonlinear feature perception abilities, enabling fast and effective decision-making, which mitigates the impacts of uncertainties in power systems [29].

However, existing DRL methods face several critical limitations that hinder their practical application. First, many methods employ multilayer perceptrons (MLPs) for their value or policy networks. By processing the power system's state as a flattened vector, these networks overlook the graph-based topology of grids. Instead of leveraging this inherent structure, MLPs are forced to infer these complex, spatially-dependent physical interactions from a one-dimensional data structure, which is usually inefficient and ineffective [30]. This impairs the agent's ability to comprehend the physical connections of power equipment, ultimately leading to poor generalization for unseen network configurations [31].

To address this challenge, recent research has introduced Graph Neural Networks (GNNs) to DRL, which can naturally process the grid's topological structure. GNNs operate via a message-passing mechanism that allows each node to aggregate information from its neighbors, thereby capturing the complex physical interdependence within the power system [32]. For instance, a scalable graph reinforcement learning (GRL) algorithm is proposed for stochastic dynamic dispatch, effectively handling topology-aware decisions [33]. A contextual meta GRL is proposed for dynamic dispatch with high generalization and few-shot adaptation capabilities [34]. A robust voltage control method is proposed that integrates physics-informed graph representation and DRL. This method effectively regulates voltage in distributed systems while simultaneously addressing challenges of anomalous measurements and model parameters [35]. However, these methods are limited to a single snapshot of the system's state. This reliance on instantaneous information overlooks crucial temporal dynamics. Consequently, the algorithm cannot leverage historical source-load data to predict future variations. This results in control actions that lack foresight and are often inadequate for mitigating the severe power fluctuations driven by these volatile source-load.

Recognizing this limitation, several recent studies have attempted to integrate temporal information into graph-based DRL frameworks. Early efforts primarily relied on stacking historical observations into a single, concatenated state vector to provide the agent with recent context. The recent voltage values are integrated into a vector for voltage stability control (VSC) in [36]. However, by only integrating historical data into the state vector without modifying the network architecture, the method struggles to effectively capture complex temporal dynamics. The work in [37] solves the corrective control tasks by stacking the last six system states to inform its topology control policy, with a Transformer as its graph network. However, its treatment of temporal information is limited; while the Transformer effectively processes spatial graph embeddings, temporal dependencies are captured simply by stacking the six most recent system states as input features. An attention-based spatial-temporal GNN is proposed for voltage control and renewable accommodation in [38], which effectively captures various temporal periodicities, such as recent, daily, and weekly, through dedicated attention mechanisms. However, in large-scale grids with high renewable penetration, unpredictable contingencies necessitate rapid, near-real-time corrective actions to resolve line violations [39]. The resulting system states are complex and highly dynamic, lacking the periodic patterns that these attention mechanisms are designed to exploit. Consequently, a gap exists in developing a spatial-temporal DRL framework specifically engineered for the dynamics and high-stakes decision-making inherent to corrective control.

To bridge this critical gap, this paper proposes a novel corrective control framework based on a Graph Attention Temporal Network and Reinforcement Learning (GATN-RL). Our method is specifically designed to overcome the limitations mentioned above by capturing the grid's spatial-temporal features to perform efficient and robust corrective actions, including both generator re-dispatch and topology adjustments. The effectiveness of this approach is validated on standard 36-node and 118-node test systems.

1. To address the inability of conventional DRL to perceive power system topology, we propose a novel GATN-RL framework for corrective control. This overcomes the trade-off between speed and accuracy in traditional methods and provides superior feature representation compared to MLP-based agents.
2. Considering the impacts of high renewable energy fluctuations, we propose a temporal information fusion mechanism. This enables the agent to capture long-term temporal dependencies and execute more precise corrective control actions.
3. To improve the agent's robustness and generalization to unseen scenarios, we develop an adaptive topological embedding. This mechanism dynamically encodes changing grid structures, ensuring effective and reliable performance when the agent confronts novel network configurations and operating conditions.

The remainder of this paper is organized as follows. Section 2 details the physical modeling of power system corrective control and its formulation as a Markov Decision Process. Section 3 provides a comprehensive explanation of the proposed GATN-RL algorithm. Section 4 presents the case studies and simulation results, followed by the conclusions.

2. Problem formulation

This section outlines corrective controls' objectives and physical constraints, subsequently formulating the problem as a Markov decision process (MDP).

2.1. Power system corrective control model

Objective: Corrective control aims to mitigate transmission line overloads while minimizing the cost of control adjustments. The objective function for this optimization problem is defined as (1):

$$\min \sum_{t=1}^{T} \left[C \left(\sum_{i=1}^{N_G} \left| \Delta P_{i,t}^{G} \right|, \Delta N_{\text{topo},t} \right) + \sum_{i=1}^{N_L} \max \left(\rho_{i,t} - \bar{\rho}_i, 0 \right) \right] \tag{1}$$

Constraints: This paper presents corrective control actions for power systems, specifically employing generator re-dispatch and bus splitting to maintain secure operation. Bus splitting reconfigures the grid by changing a power equipment's connection from one bus to another within the same substation, thereby altering power flow distribution. For instance, Fig. 1 illustrates a case where a line is experiencing an overload. This switch from Bus 1 to Bus 2 successfully alleviates congestion by rerouting power and reducing the line flow to 100 MW. All power equipment, including generators, loads, and line terminals, is connected to one of the two buses or is disconnected. This arrangement gives rise to a combinatorially large number of potential network configurations. To ensure supply reliability and meet user demand, only line terminals are permitted to be completely disconnected from both buses.

Considering the operational limits of generator capacity bounds, transmission line thermal limits, voltage stability requirements, and network security, the mathematical constraints are described as Eqs. (2)–(7):

$$\sum_{i=1}^{N_G} P_{i,t}^{G} = \sum_{j=1}^{N_L} P_{j,t}^{L} \tag{2}$$

$$P_{i,\text{Gmin}} \le P_{i,t}^{G} \le P_{i,\text{Gmax}} \tag{3}$$

$$\Delta P_{i,t}^{G} \ge \min(P_{i,\text{Gmin}} - P_{i,t}^{G}, -R_{\text{down}}) \tag{4}$$

$$\Delta P_{i,t}^{G} \le \min(P_{i,\text{Gmax}} - P_{i,t}^{G}, R_{\text{up}}) \tag{5}$$

$$v_{\min} \le v_{i,t} \le v_{\max} \tag{6}$$

$$\rho_{\max,t} \le \bar{\rho} \tag{7}$$

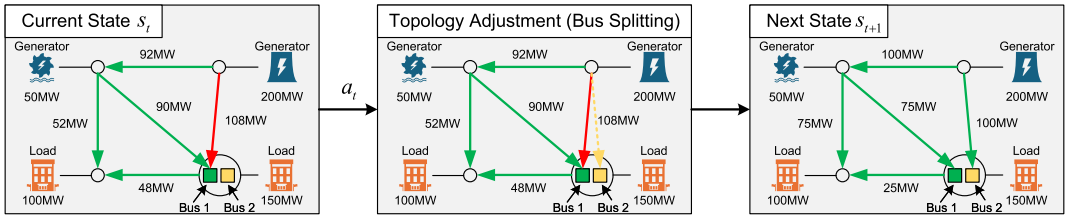


Fig. 1. The corrective control process of a double bus-bar power system.

Eq. (2) represents the power flow balance constraints. Eq. (3) represents the generator capacity constraints. Eqs. (4) and (5) denote the generator ramp constraints. Eq. (6) denotes the voltage bound constraints. Eq. (7) denotes the load ratio constraints. For computational simplicity, $\overline{\rho}_i$ is set to 1.

2.2. MDP of corrective control

In this section, we address the corrective control process using an MDP. The MDP consists of five key components: the action space $\mathcal{A}$, the state space $\mathcal{S}$, the reward function $\mathcal{R}$, the transition matrix $\mathcal{P}$, and the discount factor γ.

State space $\mathcal{S}$: The state space is designed to provide the agent with a comprehensive snapshot of the power system's operational status. It encapsulates critical operational data, beginning with power flow variables that include the active and reactive power of loads generators, and lines. Additionally, the state space incorporates power system topology states to inform network adjustments and line loading ratios, which serve as corrective control targets. The state at time t is formally defined as s_t in Eqs. (8) and (9):

$$s_t = \{P^L, Q^L, P^G, Q^G, P^O, Q^O, P^E, Q^E, c, \rho\} \tag{8}$$

$$c_i = \begin{cases} 0 & \text{disconnection} \\ 1 & \text{connection to bus 1} \quad i \in \Omega \\ 2 & \text{connection to bus 2} \end{cases} \tag{9}$$

Action space $\mathcal{A}$: The action space comprises bus splitting and redispatching generators, as represented in Eq. (10). Bus splitting eliminates line overloads by switching the connection to another bus, as shown in Fig. 1. Eq. (11) defines the bus splitting action $\mathbf{a}_B$. A transmission line can be connected to either bus 1 or bus 2, or it can be left disconnected. In large-scale power systems, the discrete and combinatorial nature of topological adjustments leads to a vast number of corrective actions. To manage this complexity, we employ Monte Carlo Tree Search (MCTS), which builds a decision tree and iteratively refines it through selection, expansion, simulation, and backpropagation, guided by the Upper Confidence Trees (UCT) algorithm to efficiently explore the most promising actions [27].

Generator re-dispatch action $\mathbf{a}_G$ is discretized into upward or downward maximum ramping bound actions, as shown in Eq. (12).

$$\mathcal{A} = (\mathbf{a}_B, \mathbf{a}_G) \tag{10}$$

$$\mathbf{a}_B = (i, c_i) \mid i \in \Omega_L, c_i \in 0, 1, 2 \cup (i, c_i') \mid i \in \Omega \neg \Omega_L, c_i' \in 1, 2 \tag{11}$$

$$\mathbf{a}_G = \{\Delta P_{i,t}^G \mid \Delta P_{i,t}^G \in \{R_{\text{up},i}, -R_{\text{down},i}\}, \forall i \in N_G\} \tag{12}$$

Reward space $\mathcal{R}$: The objective of corrective control is to alleviate line overloads using minimal control actions, thereby ensuring both the economic efficiency and operational security of the power system. Considering that the cost of bus splitting is much smaller than redispatching generators, the reward function quantifies the operational state of the power system based on line load ratios, which determine its stability. It is composed of two components: an action cost reward $r_{\text{cost},t}$ and a line overload penalty $r_{\rho,t}$, as formulated in Eqs. (13)–(15).

$$r_t = r_{\text{cost},t} + r_{\rho,t} \tag{13}$$

$$r_{\text{cost},t} = -C_P |\Delta P_t^G| - C_A N_{\text{topo},t} \tag{14}$$

$$r_{\rho,t} = \begin{cases} -\sum_{i=1}^{N_L} \max(\rho_{i,t} - \overline{\rho_i}, 0) & \exists i : \rho_{i,t} > \overline{\rho_i} \\ 1 & \forall i : \rho_{i,t} \leq \overline{\rho_i} \end{cases} \tag{15}$$

$r_{\text{cost},t}$ accounts for the operational costs of re-dispatching generators and topology adjustment. $r_{\rho,t}$ is determined based on the magnitude of $\rho_{i,t}$, which becomes negative when any $\rho_{i,t}$ exceeds $\bar{\rho}_i$, and becomes positive when all $\rho_{i,t}$ values remain below $\bar{\rho}_i$.

The transition matrix $\mathcal{P}$ is defined by Eq. (16).

$$\mathcal{P}(s'|s, a) = \Pr(s_{t+1} = s' | s_t = s, a_t = a) \tag{16}$$

$\mathcal{P}$ represents the inherent stochasticity of the environment, which is implicitly determined by the environmental dynamics.

The discount factor γ is incorporated into the reward calculation as shown in Eq. (17).

$$R_t = \sum_{k=t}^{T} \gamma^{k-t} r_k \tag{17}$$

Since $\gamma \in (0, 1)$, rewards that are temporally distant from the initial state contribute progressively less to the cumulative reward, ensuring that the DRL model prioritizes immediate rewards while appropriately balancing the influence of future rewards in the decision-making process.

3. Proposed method

This section introduces a corrective control method for power systems based on a GATN-RL framework. First, we detail the representation of graph attention networks. Then we introduce fusion mechanism that integrates GAT and Transformer networks. Next, we present the topology embedding mechanism. Finally, we describe the algorithm's training process, which utilizes demonstration learning.

3.1. Representation of graph attention networks

The power system is modeled as an undirected graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{A}_t)$, where $\mathcal{V}$ is the set of nodes representing physical power equipment, including generators, loads, line origins, and extremities. $\mathcal{E}_t$ is the set of edges representing their interconnections at time t. Considering that the focus of this study is on the network topology and connectivity, rather than the specific electrical parameters of the lines, the edges of the graph are unweighted and instead represent the physical connectivity between power system components. An edge represents the direct physical connection between two pieces of power equipment. As illustrated in the accompanying Fig. 2, components like a generator, a load, a line origin and extremity connected at a common bus are treated as directly connected nodes. $\mathbf{A}_t$ is the corresponding adjacency matrix. The adjacency matrix captures the time-varying topology of the network, as defined in Eq. (18).

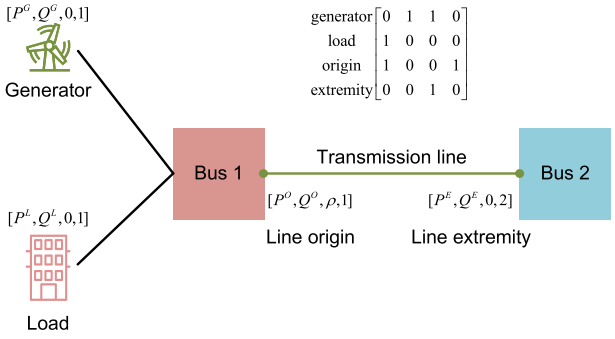


Fig. 2. The establishment of the adjacency matrix and feature matrix.

The operational state-space observations from Section 2.2 are mapped onto their corresponding nodes to form a feature matrix $\mathbf{X}_t$. Each row of $\mathbf{X}_t$ corresponds to a node $i \in \mathcal{V}$ and contains a feature vector describing its state, as formulated in Eq. (19). For equipment types that do not possess a certain attribute, the corresponding feature is padded with zero to maintain a consistent matrix structure.

$$\mathbf{A}_t[i,j] = \begin{cases} 1 & i \text{ connect to } j \\ 0 & i \text{ is disconnected to } j \end{cases} \qquad j \in \mathcal{N}(i) \tag{18}$$

$$\mathbf{X}_t = \begin{bmatrix} P^G & Q^G & 0 & c_i^G \\ P^L & Q^L & 0 & c_i^L \\ P^O & Q^O & \rho & c_i^O \\ P^E & Q^E & 0 & c_i^E \end{bmatrix} \quad i \in \Omega \tag{19}$$

We utilize a Graph Attention Network (GAT) to process graph-structured data via an attention mechanism. Unlike traditional Graph Convolutional Networks (GCNs), GATs employ multi-head attention to dynamically assign importance weights to neighboring nodes, enabling a more precise capture of complex interactions within the power system topology. The hidden representation of each node is updated using a masked self-attention strategy. To quantify the influence of neighboring nodes, attention coefficients are computed as defined in Eqs. (20) and (21). The updated representation for a target node is then derived by aggregating feature information from its neighbors, weighted by these coefficients. This graph attention convolution process is defined in Eq. (22).

$$e_{ij} = a(\mathbf{W}\mathbf{h}_i, \mathbf{W}\mathbf{h}_j) \quad j \in \mathcal{N}(i) \tag{20}$$

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(e_{ij}))}{\sum_{k\in\mathcal{N}(i)} \exp(\text{LeakyReLU}(e_{ik}))} \tag{21}$$

$$\mathbf{h}_i' = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}\mathbf{W}\mathbf{h}_j\right) \tag{22}$$

3.2. Graph attention temporal network reinforcement learning

In power systems with high renewable energy penetration, relying solely on the current system state is insufficient to capture the variability of renewable generation. Traditional Deep Reinforcement Learning (DRL) algorithms typically utilize Multi-Layer Perceptrons (MLPs) as decision networks. However, by processing inputs as one-dimensional vectors, MLPs are inherently unable to capture the topological features of a power system. These features, which define the physical connectivity and power flow relationships between components, are critical for informing effective corrective actions. Therefore, we propose a feature extraction framework based on GATN-RL. This framework expands the model's receptive field to encompass both current and historical operational data, providing a comprehensive understanding of the temporal dynamics and topological information required for robust corrective control.

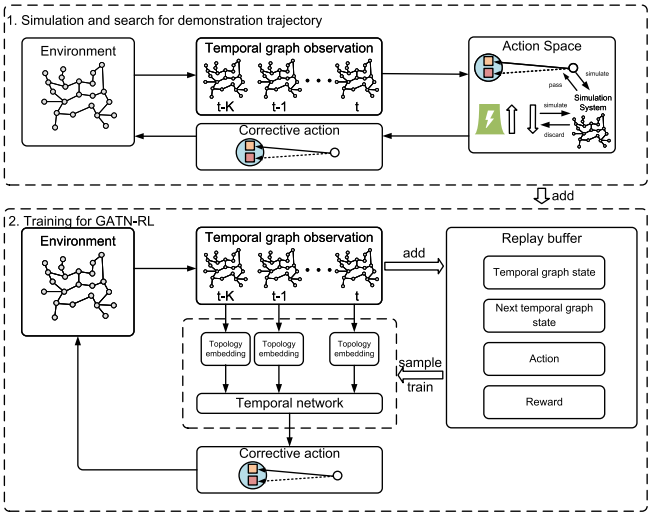


Fig. 3. The framework of the proposed method.

Fig. 3 illustrates the framework of the proposed method. Before training, high-quality control actions are generated via simulation, filtered, and stored in a replay buffer as initial trajectories. During operation, the agent extracts temporal, graph-structured power system observations from the environment. These observations are fed into the GATN, which generates corrective control actions. The application of these actions to the environment completes the control loop. Trajectories generated during this interaction are continuously added to the replay buffer. Concurrently, the agent samples data from the replay buffer to train the GATN.

The architecture of the GATN is depicted in Fig. 4. It comprises four key modules: a topology embedding module, a GAT-based state encoding module, a Transformer temporal module, and a dueling Q-network module. The topology embedding module is detailed in Section 3.3. The state encoding module consists of stacked GATs with multi-head attention, which aggregates information between power system nodes according to learned attention weights.

Based on the correlation between a node and its neighboring nodes, attention coefficients for each neighbor are calculated, producing $|\mathcal{N}(i)|$ vectors of length F through weighted summation.

The Transformer network is employed to capture the temporal dependencies within the data. Power systems with large-scale topologies and high penetration of renewable energy exhibit complex temporal dynamics, including intermittent generation and abrupt disturbances. Traditional sequential models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM), often struggle to capture the dependencies inherent in these dynamics. Their limited ability to associate the current system state with past events makes it difficult to distinguish between routine fluctuations and instability. However, the Transformer architecture overcomes this limitation by directly capturing dependencies across all time steps. Its self-attention mechanism allows the model to dynamically weigh the significance of the entire operational history, thereby effectively identifying critical temporal patterns.

To mitigate potential information loss from using average pooling alone, this work concatenates feature vectors from both average and max pooling operations, resulting in feature vectors of length $2F$. These vectors are then stacked along the temporal dimension, forming the final GAT output features as described in Eq. (23). To capture temporal dynamics, the sequence of GAT-processed feature vectors is fed into a Transformer network, as detailed in Eq. (24). The spatial-temporal

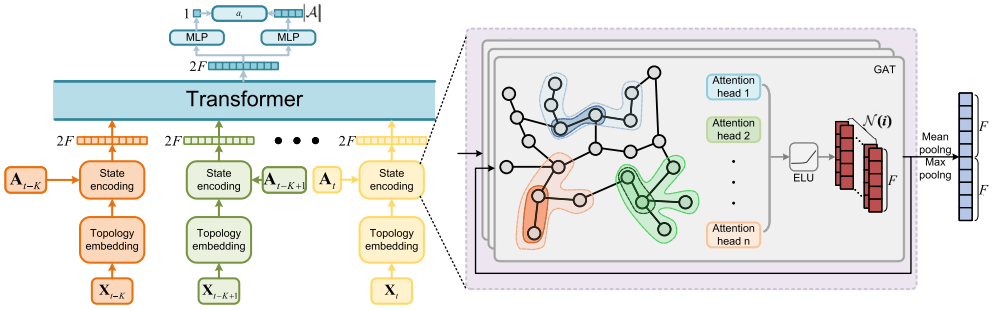


Fig. 4. The framework of GATN.

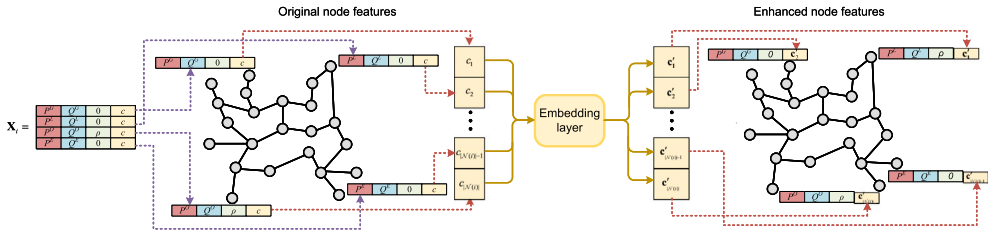


Fig. 5. The process of the topology embedding mechanism.

features extracted by the Transformer are then fed into the final layers of the network, which implement a dueling architecture to compute the state-value function and the advantage function for all feasible actions.

$$h_{\text{output},t} = \left[\frac{1}{n}\sum_{i=1}^{n} h'_{i,t}, \max(h'_{i,t})\right] \tag{23}$$

$$\begin{aligned} Z_t &= f^{\theta_S}(G_t, G_{t-1}, \ldots, G_{t-K}) \\ &= f^{\theta_T}(h_{\text{output},t}, h_{\text{output},t-1}, \ldots, h_{\text{output},t-K}) \end{aligned} \tag{24}$$

3.3. Topology embedding mechanism

While GATs are effective for processing graph structures, their training process assumes that test and training data are drawn from the same distribution. This assumption is frequently violated in power systems, where unforeseen events like line disconnections or unplanned topological reconfigurations create novel network structures not present in the training data. Consequently, GAT performance can degrade when encountering these unseen topologies, an out-of-distribution (OOD) challenge. This limitation can severely degrade the decision-making performance of associated DRL models.

Recent efforts have sought to address this generalization issue. For instance, some research simultaneously utilizes both a dynamic, learnable topology embedding and a static, physics-based embedding derived from the admittance matrix to capture both time-varying and inherent physical connectivity [40]. To address this challenge more directly, we propose a dedicated topology embedding mechanism, as illustrated in Fig. 5. This mechanism isolates the discrete connectivity state feature, denoted as c, from each node's feature vector. These discrete features are then mapped through an embedding layer to produce a dense, continuous vector representation. Finally, this continuous embedding replaces the original discrete feature c within each node's vector, yielding an enhanced feature set $\mathbf{c}'$.

The topology embedding mechanism enables the model to learn the intrinsic topological correlations within the power system. By learning the underlying patterns from various configurations, such as N-1 contingencies or uncommon bus connection states, the model can better generalize to OOD scenarios. The resulting embedding space captures the relationships between different connection patterns, allowing the model to make robust inferences by identifying similarities between known and novel topologies. Furthermore, converting discrete connection states into continuous representations mitigates data sparsity, which facilitates effective model training and improves gradient flow during optimization.

3.4. Demonstration learning

DRL methods face a significant challenge with data exploration efficiency in corrective control [41]. The large action space of possible corrective controls contains only a small proportion of effective actions. During the initial training stages, a DRL agent relies on random exploration to gather environmental feedback and optimize its decision-making strategy. In complex power system environments, the sparsity of effective actions makes this exploration process highly difficult. This inefficiency significantly prolongs training cycles and increases the risk of the model converging to a local optimum or failing to converge altogether.

To address the problems, we propose a demonstration learning training process, as detailed in Algorithm 1. First, an expert policy performs an exhaustive search of the action space via simulation to identify the optimal action that minimizes the maximum line load ratio. The resulting expert trajectories τ are then stored in a demonstration buffer, D_{demo}. Second, the data is sampled from both the demonstration buffer D_{demo} and a standard experience replay buffer D, governed by a mixing ratio β. By learning from high-quality, expert trajectories at the onset of training, this method improves the model's initial performance and significantly accelerates the convergence process.

4. Case study

In this section, we conducted a comprehensive case study comparing multiple algorithms to validate the performance of the proposed method in terms of training deployment, test set evaluation, and generalization capabilities.

4.1. Experiment setup

To enable efficient corrective control in power systems with high renewable energy penetration and dynamic network topologies, we developed a simulation environment using the grid2op platform [42].

Algorithm 1 The Process of demonstration learning.

Initialize demonstration replay buffer $\mathcal{D}_{demo} = \{\}$
for $i = 1$ to N_{demo} **do**
  collect the state s_i
  search the action space $\mathcal{A}$ and find the action a_i^* that minimizes the maximum load ratio
  execute a_i^*, and get the next state s', reward r, and transition (s_i, a_i^*, r, s')
  store the trajectory (s_i, a_i^*, r, s') to $\mathcal{D}_{demo}$
end for
Initialize the Q-network parameters θ and target Q-network parameters θ'
Initialize the replay buffer $\mathcal{D}$, sample factor β and batch size M
Initialize the soft update factor η
for $i = 1$ to N **do**
  initialize the state s_0
  for $t = 0$ to T **do**
    calculate the Q-value $Q(s_t, a_t; \theta)$
    obtain action a_t
    execute a_t, and get the next state s_{t+1}, reward r_t, and transition (s_t, a_t, r_t, s_{t+1})
    store the trajectory (s_t, a_t, r_t, s_{t+1}) in $\mathcal{D}$
    update state $s_t \leftarrow s_{t+1}$
  end for
  sample $\beta \cdot M$ random batch $\mathcal{B}_{demo}$ of trajectories from $\mathcal{D}_{demo}$
  sample $(1-\beta) \cdot M$ random batch $\mathcal{B}_{RL}$ of trajectories from $\mathcal{D}$
  calculate the loss function $L(\theta) = \left\| r_t + \gamma Q^{\theta'}(s_{t+1}, \arg\max_{a \in \mathcal{A}} Q^{\theta}(s_{t+1}, a')) - Q^{\theta}(s_t, a') \right\|^2$
  update the Q network parameters θ
  update the target Q network $\theta' \leftarrow (1-\eta) \cdot \theta' + \eta \cdot \theta$
end for

This environment incorporates numerous operational scenarios with significant renewable generation and topological flexibility, providing a diverse set of training cases. For training and testing, we employed two power systems: a modified 36-node system and the IEEE 118-node system, with their respective topologies illustrated in Figs. A.1 and A.2. The 36-node system, derived from a subset of the 118-node system, consists of 36 substations, 59 transmission lines, 22 generators, and 37 loads. The IEEE 118-node system comprises 118 substations, 186 transmission lines, 62 generators, and 99 loads. Both systems feature a double bus-bar configuration, which allows substations, transmission lines, and loads to be reassigned between the two busbars. At initialization, all elements are connected to the first bus. The simulations operate with a temporal resolution of 5 min. The complete dataset contains 2953 monthly power flow scenarios for the 36-node system and 2232 for the 118-node system. To facilitate rapid verification, each scenario is limited to a maximum of 288 operational steps, equivalent to 24 h.

The inherent uncertainty of renewable energy generation, coupled with load fluctuations, can lead to grid overloads. The objective for the trained agent is to resolve these overloads by adjusting generator active power setpoints and executing bus splitting actions. A system breakdown is triggered if either of the following conditions is met: (1) the load ratio on any transmission line exceeds 2.0, or (2) a single line remains overloaded ($\rho_{\max,t} > 1.0$) for 15 consecutive minutes (three time steps).

To validate our proposed method, we benchmark the GATN-RL framework against three algorithms. First, a conventional MILP model serves as a model-based benchmark to evaluate control effectiveness. Second, a D3QN agent with an MLP (D3QN with MLPs) represents the standard, topology-agnostic DRL approach, allowing us to quantify the benefits of our spatial modeling. Finally, a D3QN agent with a GAT (D3QN with GAT), which captures spatial but not temporal features, enables us to isolate and demonstrate the crucial contribution of our framework's temporal processing capabilities for managing dynamic grid conditions.

Table 1
Hyperparameter settings.

Method	Parameters	Value
GATN-RL (Proposed)	Hidden size of GAT	[128, 128]
	Hidden size of Transformer	[128, 128]
	Layers of Transformers	3
	Nheads of Transformer	4
	Positional encoding	Sinusoidal
D3QN with MLPs	Learning rate	10^{-4}
	Hidden size	[256, 256, 256, 256]
D3QN with GAT	Learning rate	$3{\cdot}10^{-4}$
	Hidden size	128
	Batch size	256
Shared parameters	State and action space size of 36-node	[187, 500]
	State and action space size of 118-node	[590, 1008]
	Hidden size of dueling network	[128,1]
	C_P, C_A	1, 0.2
	γ	0.95
	η	0.01
	$\bar{\rho}_i$	0.95

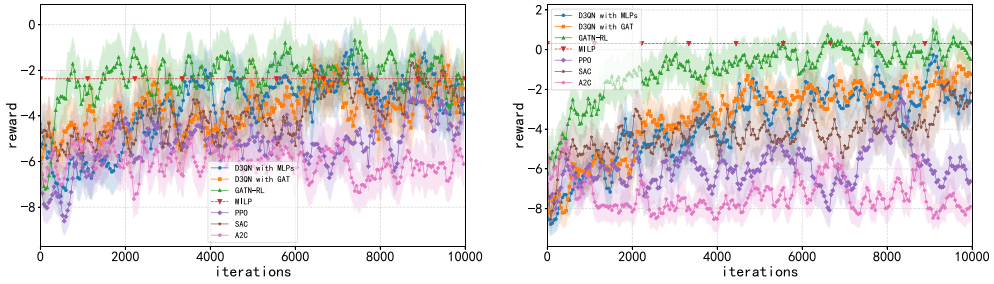
The hyperparameters are detailed in Table 1. The hyperparameters for all baseline models were carefully tuned to ensure a fair comparison. The experimental setup utilized Python 3.12.9 and PyTorch 2.6.0 (with CUDA 12.8), running on a Windows 11 system equipped with an NVIDIA RTX 4060 Ti (16 GB) GPU.

4.2. Learning performance of the proposed method

To evaluate its corrective control performance, the proposed GATN-RL method was compared to D3QN using MLP and GAT decision networks, the traditional MILP method, and three DRL algorithms: Proximal Policy Optimization (PPO), Soft Actor-Critic (SAC), and Advantage Actor-Critic (A2C). Each algorithm was trained for 10,000 episodes on both 36-node and 118-node power systems. The training progress, tracked by rewards per episode, is presented in Fig. 6.

The results demonstrate that in the 36-node system, which presents lower complexity due to its smaller scale, the proposed algorithm exhibits slight improvements in convergence performance compared to other algorithms, as shown in Fig. 6(a). However, our proposed method exhibited significantly faster and superior convergence speed. When tested on the more complex 118-node system, the increased scale and topological diversity introduced significant challenges. The three benchmark methods experienced performance degradation. As illustrated in Fig. 6(b), the increased scale and topological diversity of this larger grid pose significant challenges for the benchmark methods, all of which exhibit considerable performance degradation. In contrast, the GATN-RL agent maintains robust learning and effective corrective control. This superior performance is attributed to its ability to efficiently extract critical topological features from the grid's graph representation and accurately capture dynamic source-load fluctuations.

Fig. 7 illustrates the corrective control processes of various methods for 36-node systems. Following the line fault, the GATN-RL agent effectively mitigates the overload by applying corrective control actions. This successfully stabilizes the maximum line load ratio under the critical threshold for the entire 288-step control duration. In contrast, all baseline methods fail to maintain system stability. This performance disparity stems from architectural deficiencies in the baseline models. Traditional reinforcement learning algorithms such as PPO, A2C, SAC, and D3QN with MLPs lack a graph neural network module, which makes them insensitive to topological changes like line outages and prevents them from formulating effective corrective responses. Similarly, the D3QN with GAT model lacks a temporal processing module, hindering



(a) Rewards of 36-node systems (b) Rewards of 118-node systems

Fig. 6. The reward curve during training of algorithms on 36-node and 118-node systems.

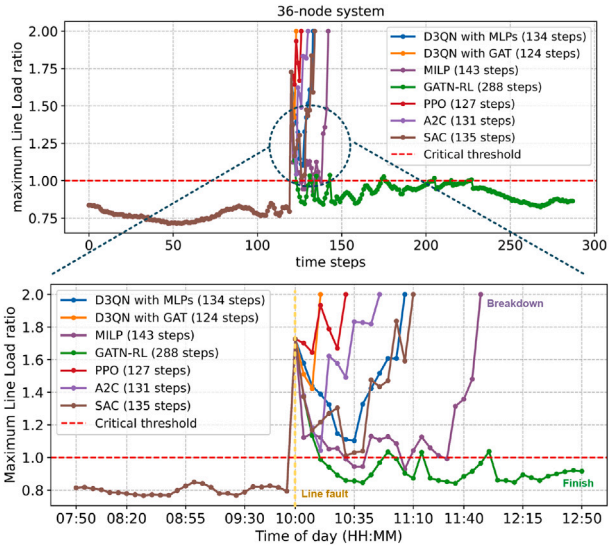


Fig. 7. The corrective control processes of all methods on a single scenario.

its ability to perceive sharp, short-term fluctuations in generation and load. This deficiency prevents it from using historical data to infer future trends and make precise control decisions. As detailed in the magnified plot, these methods are unable to contain the initial overload, resulting in severe, subsequent spikes in the line load ratio that culminate in system breakdown. This failure occurs rapidly across all baselines, with the system collapsing at 124 steps for D3QN with GAT, 127 for PPO, 131 for A2C, 134 for D3QN with MLPs, 135 for SAC, and 143 for MILP. Therefore, the proposed GATN-RL method is the only approach that ensures continuous system operation. Completing the full control sequence demonstrates good corrective control and enhances operational security through a more stable line load.

To validate the performance of the proposed method, it was tested on 200 scenarios per power system that were excluded from the training set. The comparative results are presented in Table 2. The alive steps represent the average operational steps of the test set. The overload rate is the percentage of alive steps where the system operates in an overload condition. Average actions denote the average number of corrective control actions taken per scenario. The success rate represents the percentage of scenarios completed within the test set. Average alleviation time measures the average duration from the occurrence of an overload to its alleviation through corrective control. Average decision time is the average computational time required for the model to make a decision. Finally, the cost is quantified by Eq. (14).

The results demonstrate that across both the 36-node and 118-node systems, GATN-RL achieves the highest alive steps and success rate. Its performance is nearly on par with the traditional MILP algorithm, with a difference of less than 1 %. Compared to conventional reinforcement learning methods, GATN-RL increases the success rate by 2.7 % to 25 % in the 36-node system and 9.83 % to 69.5 % in the 118-node system. Moreover, when compared against models using only MLP or GAT networks, our method shows improvements of 4.7 % and 9.2 % to 12.6 % for the 36-node and 118-node systems, respectively. This indicates that our proposed method implements more effective corrective control policies, successfully stabilizing the system for longer periods and resolving a higher percentage of contingencies than all baselines. This is because the proposed method integrates graph attention mechanisms with a temporal network, unlike MLP-based approaches without topology perception or GNN-only methods that rely on instantaneous snapshots. This architecture provides the agent with a holistic understanding of both the grid's physical structure and its dynamic evolution of source-load fluctuation, enabling more foresighted and robust decision-making that better prevents system collapse.

Furthermore, GATN-RL exhibits superior operational efficiency, as evidenced by the lowest overload rate, the shortest average alleviation time, and the lowest cost in both test systems. By understanding temporal trends in source and load, agents can select corrective actions that not only resolve immediate overloads but also preemptively mitigate future violations. This leads to a marked reduction in the time the system spends in hazardous overload conditions, minimizing the economic penalties associated with control actions and thereby preventing more severe faults.

The MILP algorithm's execution time is 2.585 s for the 36-node system and exceeds 30 s for the 118-node system. In stark contrast, all DRL-based algorithms, including GATN-RL, achieve an average decision time of under 0.03 s, which is less than 0.2 % of the MILP computation time. This disparity shows that DRL agents make decisions via a rapid forward pass through the neural network. Conversely, MILP solves a complex optimization problem for each control decision, a process whose computation time scales exponentially with system size. This latency renders MILP impractical for urgent corrective control.

4.3. The generalization of the proposed method

To rigorously assess the topological generalization capability of the proposed method, we designed an experiment to evaluate its performance in completely unfamiliar network environments. In a power grid model, reconfiguring the terminal connections of several lines fundamentally alters the network's connectivity, creating entirely new topologies, and evaluating the agent's ability to comprehend and adapt to novel network structures. To implement this, we enhanced topology diversity through a randomization process. At initialization, five transmission lines in the 36-node system and ten in the 118-node system were randomly selected. These lines were then reconfigured by switching one of their terminal connections to bus 2. Using this setup, we generated a new test set to evaluate the models, which had been trained on the original dataset.

Table 2
The experimental results on the test set.

System	Algorithm	Training time (h)	Alive steps	Overload rate (%)	Average actions	Success rate (%)	Average alleviation time (min)	Average decision time (s)	Cost
36-node	D3QN with MLPs	**2.66**	223.18	3.39	6.02	52.6	13.94	0.016	399.68
	D3QN with GAT	3.03	223.75	4.08	5.88	53.5	16.81	0.018	229.05
	GATN-RL	3.39	**233.67**	**3.38**	**5.62**	**58.8**	**10.72**	0.018	**194.07**
	MILP	/	233.15	4.01	8.92	49.5	12.77	2.585	279.59
	PPO	3.35	214.67	3.93	5.76	42.0	16.55	0.021	237.12
	SAC	3.02	227.01	3.73	5.64	53.5	15.15	**0.012**	219.10
	A2C	3.61	186.66	4.28	5.21	21.5	15.62	0.019	201.33
118-node	D3QN with MLPs	**7.08**	230.97	3.84	6.44	61.5	14.80	0.020	789.42
	D3QN with GAT	8.12	238.25	4.28	6.38	62.5	10.66	**0.016**	741.50
	GATN-RL	8.32	**260.24**	**3.59**	**6.22**	**80.5**	**16.13**	0.027	**739.37**
	MILP	/	258.38	5.28	8.48	77.9	12.15	30.475	762.10
	PPO	8.41	236.93	5.06	6.37	64.5	13.53	0.027	766.23
	SAC	8.11	234.61	3.68	6.29	72.3	10.90	0.026	763.27
	A2C	8.53	153.51	7.17	7.67	44.0	16.96	0.028	772.64

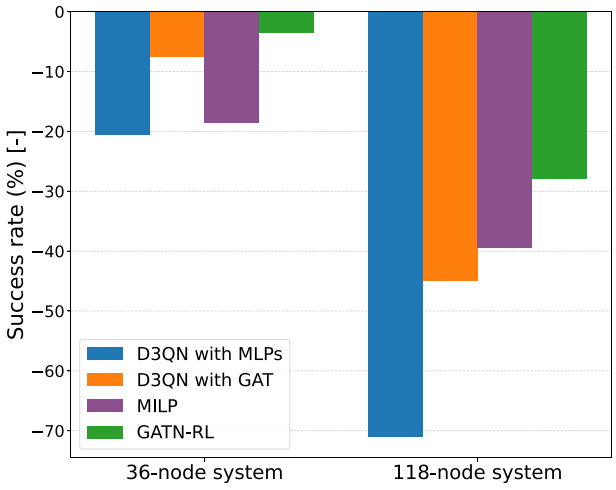


Fig. 8. Reduced success rate using a randomized initial topology.

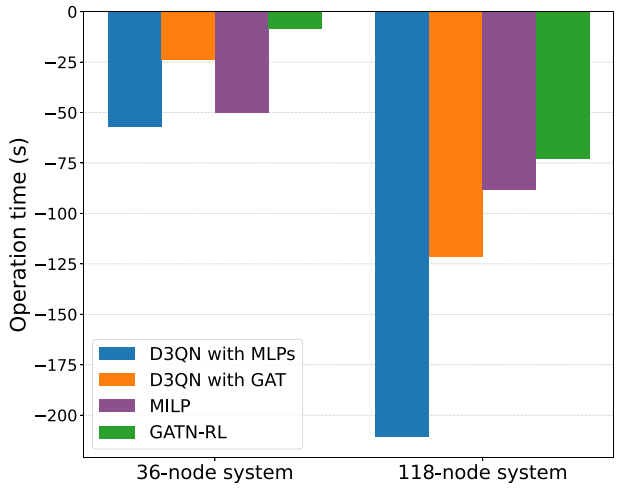


Fig. 9. Reduced operation time using a randomized initial topology.

Figs. 8 and 9 illustrate the reduced success rate and operation time achieved with a randomized initial topology. The figures show that all methods experienced varying performance degradation under random initial topology conditions. In the 36-node system, which has relatively lower topological complexity, all methods exhibited operation time losses of less than 5 h. While the baseline GAT showed some resilience in the simpler 36-node system, its limitations became apparent in the large-scale 118-node system. In contrast, our proposed GATN-RL model integrates a topology embedding mechanism that enables the model to learn a more generalized and compositional topological representation. When tested on novel topologies, the proposed method exhibits only a minor degradation, with a 9.85 % reduction in operation time and a 4.90 % decrease in success rate.

In the 118-node system, due to increased topological scale and randomly switched buses, both MLPs and GAT experienced substantial performance degradation, with operation time reductions of 17.5h and 10.2h, respectively. While MILP can fully account for system topology changes through physical modeling and solution techniques, it requires enormous computational resources, as shown in Table 2. By contrast, the proposed algorithm not only extracts topological information using graph neural networks but also employs the topology embedding mechanism to capture bus connection information for individual electrical elements further. Compared to the proposed method's performance drop, the MLP-based agent's operation time fell by an additional 66.23 % and its success rate by an additional 60.57 %. Similarly, the GAT-based agent's operation time and success rate fell by an additional 60.01 % and 37.78 %, respectively, highlighting the effectiveness of the topology embedding mechanism.

To evaluate the trained agent's adaptability to renewable energy, its performance is assessed under two scenarios with distinct levels of renewable energy uncertainty, as shown in Fig. 10. Scenario 1 has a high renewable energy penetration rate level, whereas Scenario 2 features a steep decline in penetration between time steps 100 and 150. For each scenario, the left panels depict the renewable energy penetration rate, while the right panels show the corresponding maximum line load ratio resulting from the agent's corrective control actions.

In Scenario 2, while the MILP and D3QN with GAT models successfully manage the initial overload event because of renewable energy fluctuation around timestep 100 to 150, their reactive control leaves them vulnerable to subsequent, more severe disturbances. The GATN-RL model utilized its Transformer-based architecture to analyze historical data windows, enabling it to recognize patterns of volatility in renewable energy and apply preemptive adjustments. This foresight allows the GATN-RL model to maintain stability through anticipatory actions during the overload at timestep 215. When a line disconnection causes a major overload at timestep 240, the GATN-RL model exhibits a lower maximum line load ratio compared to the other two algorithms. This reduced system stress consequently makes it easier to alleviate the line overload via corrective control.

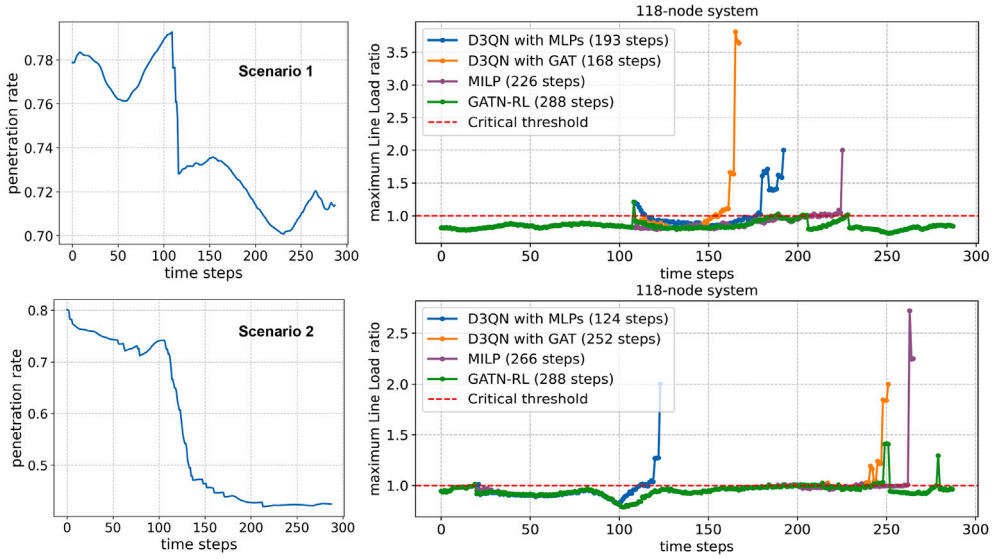


Fig. 10. The corrective control processes with variable renewable energy penetration ratio.

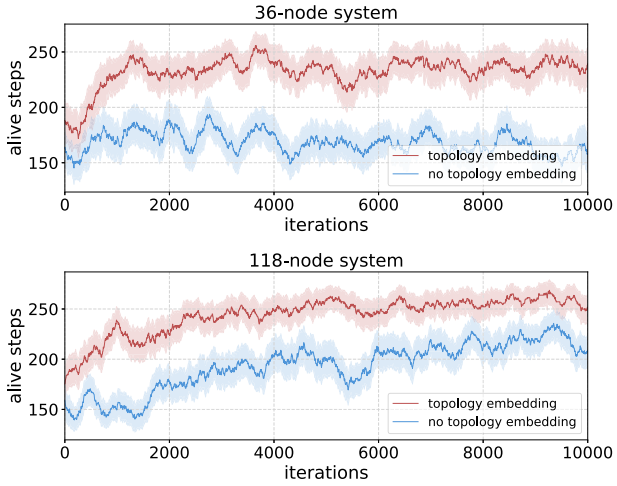


Fig. 11. Alive steps curves during training with and without topology embedding.

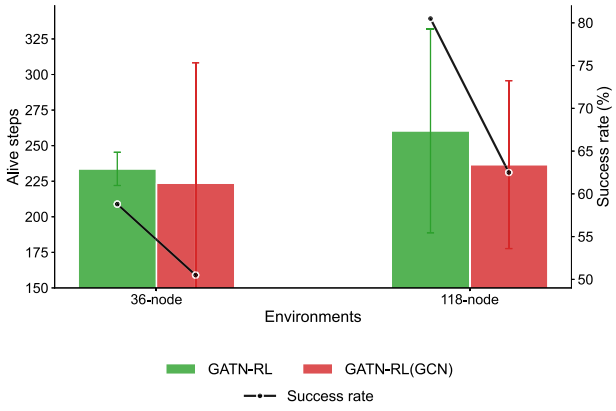


Fig. 12. Alive steps and success rate between different graph neural networks.

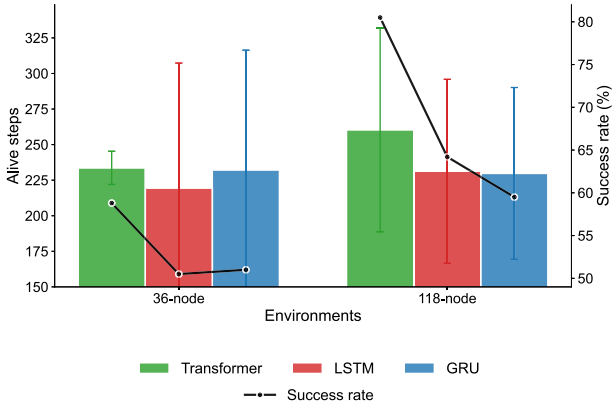


Fig. 13. Alive steps and success rate among different temporal networks.

4.4. Ablation experiments

To evaluate the effectiveness of topology embedding, we conducted experiments with and without the topology embedding mechanism while maintaining consistent hyperparameters during training. The results are illustrated in Fig. 11. In the 36-node system, incorporating topology embedding showed modest performance improvements. In the 118-node system, the integration of topology embedding yielded substantially better results, with significantly higher alive steps both during training and at convergence compared to the non-embedded method. This performance gap can be attributed to larger power systems' inherently more complex topological structures. Without topology embedding, the model struggles to extract sufficient topological features from such complex systems, which significantly hinders its learning process and leads to suboptimal performance.

To validate the effectiveness of the architectural components in the proposed algorithm, ablation studies were conducted on the Graph Attention Network (GAT) and Transformer modules. The GAT module was compared with GCN to assess its spatial feature extraction capability, as shown in Figs. 12 and 13. The results show that the GAT-based model achieves higher alive steps and success rates than the GCN variant in both the 36-node and 118-node systems. The results indicate that GAT more effectively captures the complex topological features of power grids across different scales. For the temporal analysis, the Transformer module was compared with Long Short-Term Memory (LSTM) and Gated

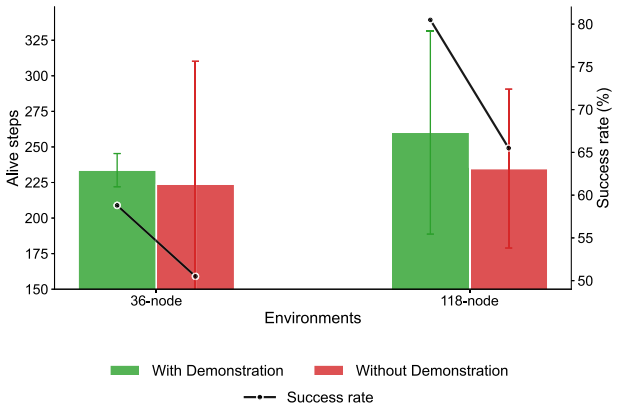


Fig. 14. Alive steps and success rate whether demonstration learning is used.

Recurrent Unit (GRU) networks. The Transformer-based model consistently outperforms both recurrent architectures, demonstrating its ability to capture temporal dependencies in power system source-load data, which is crucial for making effective, foresighted control decisions.

Furthermore, the impact of demonstration learning on model performance was tested by comparing agents trained with and without expert trajectories. The results are illustrated in Fig. 14. The agent trained with demonstrations achieved substantially higher alive steps and success rates across both test systems. This result confirms that demonstration learning effectively bootstraps the training process, guiding the agent out of inefficient initial exploration and accelerating its convergence toward a robust and high-performing control policy.

5. Conclusion

A novel Graph Attention Temporal Network with Reinforcement Learning (GATN-RL) algorithm is proposed in this paper for the power system corrective control. The proposed method effectively captures spatial-temporal dependencies in power systems by employing a GAT to process topological information and a Transformer to model temporal dynamics. To address the out-of-distribution challenge posed by unseen grid configurations, we designed a dedicated topology embedding mechanism that enhances the model's generalization capabilities. Furthermore, the training process is significantly accelerated and stabilized through demonstration learning, which overcomes the exploration inefficiency common in complex control tasks. Case studies on various power systems demonstrate that the GATN-RL method surpasses both traditional optimization methods and standard DRL approaches in corrective control performance, achieving superior results with a fraction of the computational time. The algorithm also shows strong generalization to unseen grid topologies and high variability in renewable energy generation, owing to its adaptive topology embedding and temporal feature extraction capabilities.

While the GATN-RL architecture is more complex than conventional DRL models, this design is essential for accurately representing the grid's spatio-temporal characteristics. The empirical results confirm that this complexity does not hinder real-time deployment, as evidenced by its fast decision-making time. Nevertheless, certain limitations present opportunities for future work. The current model relies on a fixed-length history of operational data, which could be improved. Future research will focus on developing methods for selectively filtering temporal information to enhance decision-making and computational efficiency. Additionally, the control flexibility could be increased by advancing beyond the current discretized action space. We aim to develop an integrated approach that seamlessly combines discrete topological control with continuous generator re-dispatch adjustments. Finally, applying and scaling the proposed framework to larger, real-world power systems remains a key avenue for future investigation.

CRediT authorship contribution statement

Haoran Zhang: Writing – original draft, Visualization, Validation, Software, Methodology. **Yiguo Guo**: Resources, Funding acquisition, Data curation. **Zhiwei Wang**: Funding acquisition, Formal analysis. **Peidong Xu**: Writing – review & editing, Resources, Project administration, Methodology. **Yuxin Dai**: Visualization, Software, Data curation. **Huaiguang Jiang**: Methodology, Formal analysis, Conceptualization. **Jun Zhang**: Writing – review & editing, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. The information about environments

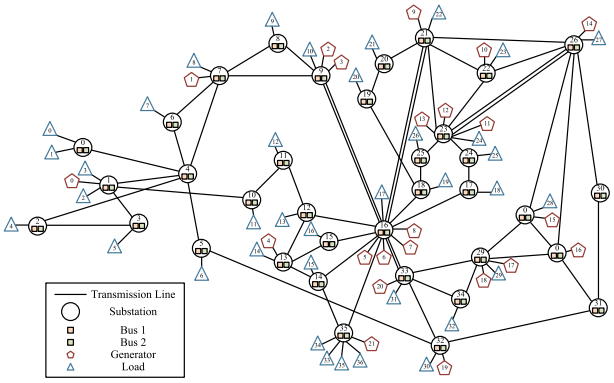


Fig. A.1. The topology of the 36-node system.

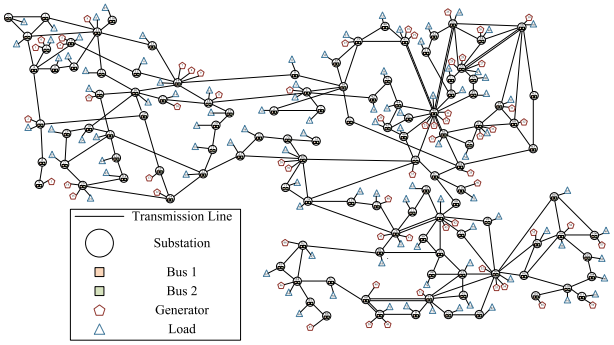


Fig. A.2. The topology of the 118-node system.

Data availability

No data was used for the research described in the article.

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